AWS Lambda Lab Exercises

1. Create Your First Lambda Function

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Create a Python Lambda function named hello-lambda. Deploy it and return 'Hello from AWS Lambda!'. Test it using the Test feature and verify the output.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

2. Read Event JSON

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Create test events containing name and course. Read the values from the event object and print a personalized greeting. Test with multiple JSON inputs.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

3. Use Environment Variables

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Create environment variables GREETING and INSTITUTE. Read them inside the Lambda function using Python's os module and combine them with the event values. Change the variable values without changing the code and test again.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

4. Calculator

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Accept two numbers in the event JSON and return addition, subtraction, multiplication and division in JSON format.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

5. String Operations

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Accept a string from the event JSON and perform uppercase, lowercase, reverse, character count and word count.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

6. CloudWatch Logs

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Add print statements to log the received event, processing steps and final output. Execute the function and verify the logs in CloudWatch.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

7. IAM Execution Role

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Open the Lambda execution role from the Configuration tab. Identify the attached policies and explain why the role is required.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

8. Start an EC2 Instance

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Create an EC2 instance and stop it. Add an IAM policy allowing ec2:StartInstances to the Lambda execution role. Pass the instance ID in the event JSON and start the instance using boto3. Verify the instance state changes to Running.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

9. Stop an EC2 Instance

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Extend the IAM role with ec2:StopInstances permission. Modify the Lambda function to stop the instance based on the event JSON. Verify the instance state changes to Stopped.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.

10. EC2 Management Mini Project

Objective: Learn the specified AWS Lambda concept through hands-on practice.

Steps: Create one Lambda function that accepts action=start or action=stop and an instance_id. Depending on the action, start or stop the EC2 instance. Handle invalid actions gracefully and log all operations to CloudWatch.

Deliverable: Submit screenshots, Lambda code, test events, and outputs for verification.